

Jacob Peyser

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Education

The Cooper Union for the Advancement of Science and Art

Bachelor of Engineering in Mechanical Engineering

Cumulative GPA: 3.78/4.00

Expected Graduation: May 2027

Notable Projects:

- **Accelerometer Telemetry:** Designed a wireless telemetry system using a Raspberry Pi Pico and WiFi-based data acquisition pipeline (Pico → WiFi router → PC) for 3-axis flight data with a calibration protocol for sensor bias, orientation determination, and axis scaling.
- **Controller Design:** Designed a lead-lag compensator in MATLAB Simulink to stabilize a motorized disc voltage input under rise-time, disturbance, and steady-state error constraints.
- **Beam Design:** Optimized a structural beam under geometric & material constraints using Wolfram Mathematica, achieving the highest failure load-to-weight ratio among 20+ students.

Honors & Research:

- **Tau Beta Pi, Engineering Honor Society:** Top 12.5% of Cooper juniors/seniors (2025)
- **Class of 1907 Award:** Annual award recognizing Cooper's top student in calculus (2026)
- **Dean's List:** All Semesters (Fall 2023 – Spring 2026)
- **Math Research Award:** Cooper Math Department Summer Stipend (2024): Numerically solved and animated PDEs via the Forward Euler Method using Wolfram Mathematica.
- **Physics Research:** Cooper (2026): Simulated the fluid dynamics of vortexes, and developed their relation to photon rings around black holes in General Relativity.

Notable Coursework:

- **Completed:**
 - **Applied Engineering:** Feedback Control, Signal Processing, CFD, Circuits, Vibrations
 - **Applied Physics:** Dynamics, E&M, Mat-Sci, Solid Mechanics, Fluids, Heat Transfer
 - **Math:** Abstract Linear Algebra, Differential Geometry, Complex Analysis, ODEs
- **Currently Enrolled:** Senior Capstone, PDEs, Manufacturing, Advanced Thermo

Professional Experience

Alpal Precision Machining Co. - Intern

June–August 2023, January 2024

- Conducted aircraft component tolerance inspections, ensuring dimensional conformance.
- Operated CNC mills and refined post-processing (sanding, deburring) to tolerance specs.
- Designed facility floor plans for the installation and integration of new machinery.
- Analyzed & documented draft schematics of aluminum components in Excel.

Tal Ron Home Improvement - General Contracting Apprentice

June–August 2021

- Assisted in home and office renovation, including vinyl floor tiling, sheetrock, and PVC piping.
- Prepared painting surfaces via proper sanding & taping techniques, followed by painting.
- Conducted floor demolition (tiling/carpeting) and ensured thorough cleanup.

Technical Skills

- Computer Aided Design: Ansys Fluent Workbench, Autodesk Fusion 360, Inventor, AutoCAD
- Programming: MATLAB (Control System & DSP), Wolfram Mathematica, Python, C++
- Microsoft Office Suite: Excel, PowerPoint, Word
- Controls & Signal Processing: Feedback Optimization, Digital Filter Design, Modal Analysis